

InvestigateIQ

AI-Powered AML Investigation Copilot

Helping AML investigators move from alert to evidence-based conclusion faster, with AI

1. Executive Summary

AI-Powered AML Investigation Copilot is an AI-assisted solution that helps financial crime investigators investigate AML alerts by bringing together customer context, transaction activity, counterparties, historical behavior and supporting evidence into a single investigation workflow.

The Copilot guides investigators through the investigation journey — **understand the alert** → **validate transactions** → **establish customer context** → **identify anomalies** → **investigate counterparties** → **resolve gaps** → **reassess** → **document the conclusion**.

AI accelerates evidence gathering, identifies relevant patterns, highlights gaps and generates an evidence-grounded investigation narrative, while the **investigator retains decision authority** over escalation, closure or reporting.

2. Business Problem & Industry Context

AML investigations are fundamentally **evidence-intensive and investigative**, but the information required to reach a conclusion is often distributed across multiple sources.

Investigators must determine:

- Why was the alert generated?
- Is the transaction activity actually unusual for this customer?
- Who are the relevant counterparties?
- What explains the observed behavior?
- What evidence supports or contradicts the concern?
- What information is still missing?
- Does the concern remain after investigation?
- How should the conclusion be documented?

The opportunity is to move from a **manual, investigator-driven information search** to an **AI-assisted, evidence-driven investigation process**.

3. Current-State Investigation Challenges

Fragmented investigation

Customer, account, transaction, KYC/CDD, alert and supporting evidence may reside across different systems and sources.

High manual effort

Investigators spend significant time gathering, reconciling and reviewing information before they can focus on actual analysis.

Complex transaction analysis

Identifying unusual transaction patterns, sequencing, counterparties and relationships can require extensive manual review.

Context switching

Investigators move between systems and documents to build a complete picture of the customer and activity.

Inconsistent investigations

The depth and structure of investigations can vary depending on investigator experience and how information is interpreted.

Documentation burden

Investigators must create a clear, chronological, evidence-based narrative that another reviewer can understand and validate.

Unresolved gaps

Important unanswered questions can remain buried in investigator notes rather than being systematically converted into next investigative actions.

4. Vision: AML Investigation Co-pilot

In Scope

Investigation Intelligence

- Alert and scenario understanding
- Customer/profile context
- Transaction analysis
- Actual vs. expected behavior comparison
- Transaction pattern identification
- Counterparty analysis
- Evidence aggregation and cross-referencing
- Identification of investigation gaps/red flags
- Recommended next investigative actions

GenAI Copilot

- Natural-language investigation Q&A
- Evidence-grounded explanations
- Investigation timeline
- Key finding summarization
- Investigator notes assistance
- Draft investigation narrative
- Evidence/source traceability

Responsible AI

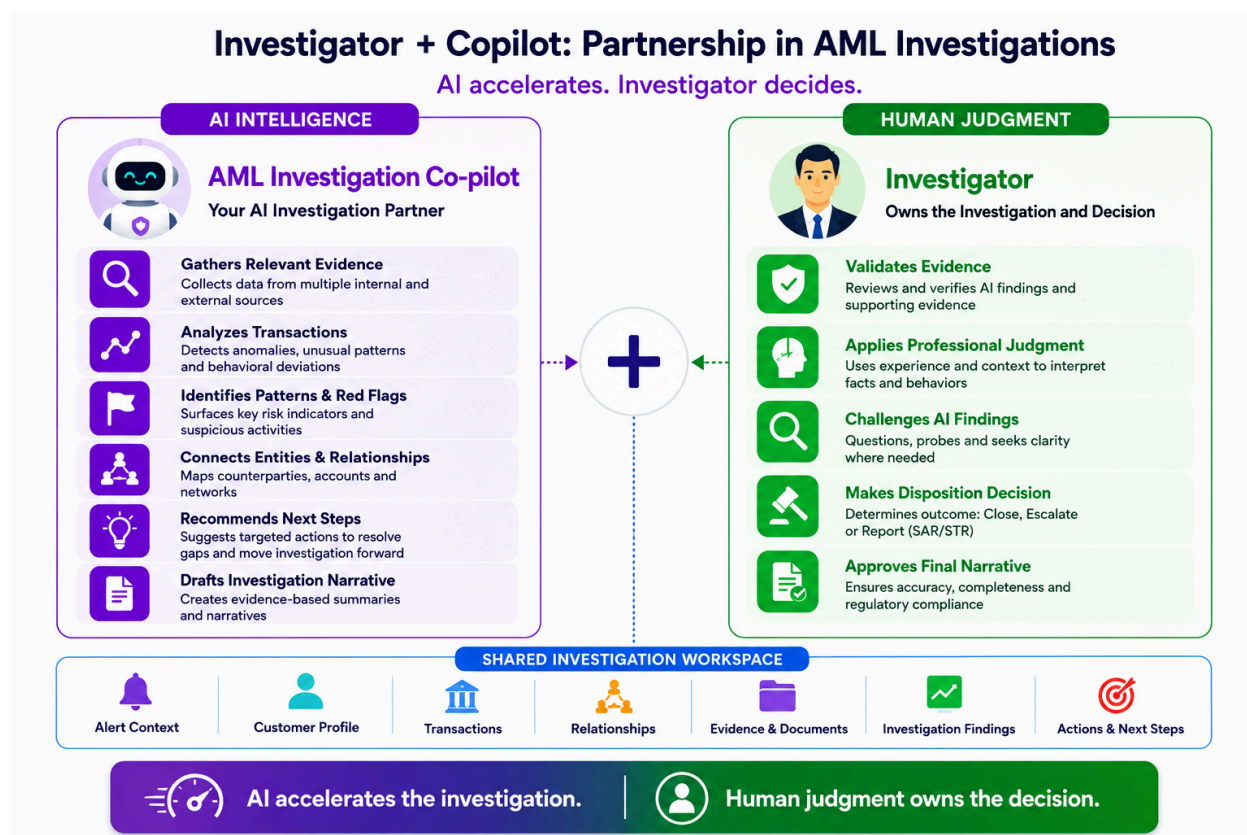
- Human-in-the-loop decision making
- Confidence indicators
- Explainability
- Source attribution
- Audit trail
- Protection against unsupported AI conclusions

Out of Scope

- Autonomous AML case disposition
- Autonomous SAR/STR filing
- Replacing the bank's AML monitoring/rules engine
- Making a final determination that money laundering occurred
- Automated customer communication
- Autonomous regulatory reporting
- Building a complete AML transaction-monitoring platform

5. Target Operating Model

Target Operating Model — Human + AI



6. AML Investigation Framework

From Alert to Investigation Conclusion

Stage	Investigation Activities	AI Role
1. CONTEXTUALIZE	Alert + customer context	Explain why alert was triggered
2. DISCOVER	Transactions + behavior + relationships	Surface patterns and connections
3. VALIDATE	Evidence + gaps + explanations	Connect findings to evidence
4. CONCLUDE	Reassess + next steps + narrative	Recommend actions and draft conclusion

7. AI Capabilities & Intelligence Layer

7.1 AI Capability Model

Investigation Stage	AI Capability	What AI does
1. Contextualize	Context Intelligence	Brings together alert, customer, account and historical information
2. Discover	Pattern Intelligence	Identifies unusual behavior, transaction patterns and relationships
3. Validate	Evidence Intelligence	Connects findings to evidence and identifies gaps
4. Conclude	Decision Support Intelligence	Synthesizes findings and drafts an evidence-grounded narrative

7.2 Core AI capabilities

1. Natural Language Understanding

- Understand investigator questions in natural language
- Translate questions into relevant data searches
- Summarize complex investigation information

2. Transaction Intelligence

- Analyze transaction volumes, values, velocity and sequencing
- Compare current activity with historical behavior
- Identify significant deviations

3. Relationship Intelligence

- Identify relationships between customers, accounts and counterparties
- Surface connected entities
- Identify potential transaction networks

4. Evidence Intelligence

- Retrieve relevant KYC/CDD documents
- Connect findings to source evidence
- Identify supporting, contradictory and missing evidence

5. Investigation Reasoning

- Identify potential red flags
- Connect multiple signals
- Highlight unresolved questions
- Recommend targeted investigation actions

6. Generative AI

- Generate investigation summaries
- Explain findings in natural language
- Draft case narratives
- Answer investigator questions

7.3 AI output principle

8. Data & Evidence Architecture

8.1 Data sources

Customer

- Customer profile
- Customer type
- Occupation/business
- Geography
- Risk rating
- Expected activity

Accounts

- Account relationships
- Account ownership
- Products
- Account status

Transactions

- Date/time
- Amount
- Currency
- Direction
- Channel
- Counterparty
- Location

Alerts & Cases

- Alert type
- Triggering rule/model
- Triggering transactions
- Previous alerts
- Previous cases
- Historical disposition

External / Supporting Evidence

- KYC/CDD information
- Invoices
- Contracts
- Statements
- Correspondence
- Other supporting documentation

Investigation History

- Investigator questions
- Previous actions
- Responses
- Findings
- Notes
- Disposition

8.2 Intelligence layer

Don't position the Copilot as simply sitting on top of raw data.

Show an intermediate **AML Intelligence Layer**:

Raw Data → Intelligence Layer → Copilot

The intelligence layer can create:

- Customer behavioral profile
- Expected vs actual activity
- Transaction patterns
- Counterparty relationships
- Network relationships
- Historical investigation context
- Risk indicators
- Evidence relationships

This is important because your capstone becomes an **AI-enabled intelligence architecture**, rather than simply “put an LLM on top of AML data.”

8.3 Evidence traceability

Every material AI-generated finding should point back to its source.

For example:

Finding: Customer received ₹5M from a previously unseen corporate counterparty.

Then:

Evidence

- Transaction ID: TXN-10234
- Date: 14-Aug-2026
- Amount: ₹5M
- Counterparty: ABC Trading
- Customer's previous transaction pattern: <₹500K/month

9. Human-in-the-Loop & Responsible AI

9.1 Human vs AI responsibility

AI Copilot	Investigator
Gather information	Validate information
Analyze transactions	Apply professional judgment
Identify patterns	Challenge AI findings
Surface relationships	Determine relevance
Identify evidence gaps	Resolve investigation gaps
Recommend next actions	Decide what action to take
Draft narrative	Approve final narrative
Explain findings	Make final disposition

9.2 Responsible AI principles

Explainability

- Why did AI identify this pattern?
- What evidence supports it?

Traceability

- Where did the information come from?
- Which transactions/documents support the finding?

Human oversight

- Investigator must validate material findings.
- AI cannot independently close or escalate a case.

Confidence & uncertainty

- AI should distinguish high-confidence findings from uncertain observations.

Bias & fairness

- Monitor whether AI disproportionately surfaces certain customer groups, geographies or profiles.

Data privacy

- Protect customer and transaction information.
- Apply appropriate access controls.

Auditability

- Maintain an auditable record of AI-generated findings, evidence and investigator actions.

9.3 Guardrails

Green — AI can automate

- Summarization
- Data retrieval
- Information organization
- Narrative drafting

Amber — AI recommends

- Risk indicators
- Investigation gaps
- Next-best actions
- Relationship insights

Red — Human decides

- Case disposition
- Escalation
- SAR/STR decision
- Final investigation conclusion

10. Target Investigator Experience

10.1 Current experience

Alert → Search multiple systems → Review transactions → Research customer → Investigate counterparties → Find documents → Identify gaps → Write narrative → Decide

The problem is fragmentation and manual effort.

10.2 Future experience

Alert → AML Investigation Copilot → AI assembles context → Investigator asks questions → AI surfaces evidence/patterns → Investigator validates → AI identifies gaps → Investigator decides

Prototype and Demo

The Rapid-Movement Business Account

1. The customer

Customer: Apex Global Trading Ltd.

Business: Import/export of industrial equipment

Customer risk rating: Medium

Expected monthly transaction volume: ₹8–12M

Primary geography: India

Typical counterparties: 10–15 established suppliers/customers

The customer has operated normally for 3 years.

2. The AML alert

The transaction-monitoring system generates an alert:

Alert: Unusual transaction activity / rapid movement of funds

Period: Last 14 days

Trigger: Transaction volume increased significantly above historical behavior.

During the 14-day period:

Transaction	Amount	Direction	Counterparty
T001	₹4.8M	In	NewCo Imports
T002	₹3.2M	In	NewCo Imports
T003	₹4.5M	Out	Orion Consulting
T004	₹2.9M	Out	Delta Trading
T005	₹3.8M	Out	Orion Consulting
T006	₹2.5M	Out	Delta Trading

Context:

₹8M comes in from a new counterparty → most of it leaves the account within 24–48 hours.